WORKED SOLUTION

# STEP III 1988, Question 1

Pure • Curve Sketching and Bounds

**STEP III 1988, Question 1**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q1

## The key idea

For the sketch, the factor $x^2e^{-x}$ is never negative, so the sign of $y$ is decided entirely by $1+x$. That, together with one derivative and the behaviour at the ends and near $x=-1$, gives the whole shape. For the integral, the cleanest move is to divide $x^2$ by $1+x$: most of the integrand then integrates to nothing, and what is left is easy to trap between two simple bounds.

## Solution

### The graph

Write $f(x)=\dfrac{x^2e^{-x}}{1+x}$, defined for all $x\neq -1$.

**Sign and zeros.** Since $x^2\geq 0$ and $e^{-x}>0$, we have $f(x)<0$ for $x<-1$ and $f(x)\geq 0$ for $x>-1$, with $f(x)=0$ only at $x=0$. The factor $x^2$ means the curve touches the $x$-axis at the origin rather than crossing it.

**Asymptotes and ends.**

- As $x\to -1$ the numerator tends to $e>0$ and the denominator to 0, so $x=-1$ is a vertical asymptote, with $f\to -\infty$ as $x\to -1^-$ and $f\to +\infty$ as $x\to -1^+$.
- As $x\to +\infty$, $f(x)\approx xe^{-x}\to 0^+$, because exponentials beat powers. So $y=0$ is an asymptote on the right, approached from above.
- As $x\to -\infty$, $\dfrac{x^2}{1+x}\approx x\to -\infty$ and $e^{-x}\to +\infty$, so $f(x)\to -\infty$, very fast.

**Stationary points.** By the quotient rule,

$$f'(x) = \frac{(2x - x^2)\mathrm{e}^{-x}(1+x) - x^2\mathrm{e}^{-x}}{(1+x)^2} = \frac{\mathrm{e}^{-x}\left(2x + x^2 - x^3 - x^2\right)}{(1+x)^2}$$
$$= \frac{x\left(2 - x^2\right)\mathrm{e}^{-x}}{(1+x)^2}.$$

So $f'(x) = 0$ at $x = 0$ and $x = \pm\sqrt{2}$. The factors $\mathrm{e}^{-x}$ and $(1+x)^2$ are positive, so the sign of $f'$ is the sign of $x(2 - x^2)$:

| interval | $x < -\sqrt{2}$ | $-\sqrt{2} < x < -1$ | $-1 < x < 0$ | $0 < x < \sqrt{2}$ | $x > \sqrt{2}$ |
|---|---|---|---|---|---|
| $f'(x)$ | $+$ | $-$ | $-$ | $+$ | $-$ |

This gives:

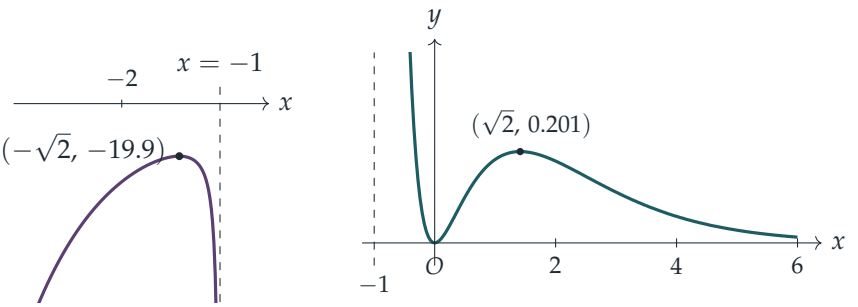
- a local maximum at $x = \sqrt{2}$, where $y = \dfrac{2\mathrm{e}^{-\sqrt{2}}}{1+\sqrt{2}} = 2(\sqrt{2}-1)\mathrm{e}^{-\sqrt{2}} \approx 0.201$;
- a local minimum at the origin;
- a local maximum at $x = -\sqrt{2}$ on the left-hand branch, where $y = \dfrac{2\mathrm{e}^{\sqrt{2}}}{1-\sqrt{2}} = -2(\sqrt{2}+1)\mathrm{e}^{\sqrt{2}} \approx -19.9$.

? **Ask yourself** How can a branch lying entirely below the axis have a maximum?

On $x < -1$ the curve comes up from $-\infty$ as $x \to -\infty$ and goes back down to $-\infty$ at the asymptote $x = -1$, so it must turn somewhere in between. The highest point is at $x = -\sqrt{2}$, and it is still well below the axis. The branch is an upside-down "hump" hanging beneath the axis, with height about $-19.9$.

On the right, the curve falls from $+\infty$ just to the right of $x = -1$ down to 0 at the origin, rises to its small maximum at $x = \sqrt{2}$, then decays to 0. It is concave up near the origin and concave down at the maximum, so there are points of inflection on either side of $x = \sqrt{2}$ (numerically near $x \approx 0.35$ and $x \approx 2.5$). We don't need their exact positions for a sketch.

The two branches have very different vertical scales, so we draw them separately. On the left, $y$ runs down to about $-70$. On the right, the $y$-scale is stretched so that the maximum of 0.2 is visible.




**The integral**

Let $I = \displaystyle\int_0^\infty \frac{x^2 e^{-x}}{1+x}\,dx$. Dividing $x^2$ by $1+x$,

$$\frac{x^2}{1+x} = x - 1 + \frac{1}{1+x}.$$

Now $\displaystyle\int_0^\infty x e^{-x}\,dx = 1$ (integrating by parts with $u = x$) and $\displaystyle\int_0^\infty e^{-x}\,dx = 1$, so the first two terms contribute $1 - 1 = 0$, and

$$I = \int_0^\infty \frac{e^{-x}}{1+x}\,dx.$$

(All three integrals converge, so splitting the integral up is legitimate.) For $x > 0$ we have $0 < \dfrac{1}{1+x} < 1$, so

$$0 < \frac{e^{-x}}{1+x} < e^{-x} \qquad (x > 0).$$

The inequalities are strict on the whole interval, and the functions are continuous, so integrating keeps them strict:

$$0 < I < \int_0^\infty e^{-x}\,dx = 1.$$

So $I$ lies between 0 and 1, as required.

**Another way.** We can skip the division. For $x > 0$, $\dfrac{x^2}{1+x} < \dfrac{x^2}{x} = x$, so $0 < f(x) < x e^{-x}$ and $0 < I < \int_0^\infty x e^{-x}\,dx = 1$. The division is worth knowing anyway, because it shows exactly where $I$ comes from.

## Checks

- Differentiating $f$ numerically at a thousand random points agrees with $f'(x) = x(2 - x^2)e^{-x}/(1+x)^2$.
- Simpson's rule on $[0, 60]$ gives $I \approx 0.59635$, and $\int_0^\infty \frac{e^{-x}}{1+x}\,dx$ gives the same value, comfortably between 0 and 1.

- A rough estimate from the sketch: on $x > 0$ the curve is a hump of height about 0.2 that has mostly died away by $x = 6$. A triangle of that height and base has area about 0.6, consistent with 0.596.

## Common slips

- Missing the stationary point at $x = -\sqrt{2}$ because it lies on the "other" branch, or calling it a minimum because $y$ is negative there.
- Saying the curve crosses the axis at the origin. The double factor $x^2$ makes it touch.
- Claiming $I < 1$ from $\mathrm{e}^{-x} \leq 1$ alone. That bound gives $\int_0^\infty \frac{x^2}{1+x}\,\mathrm{d}x$, which diverges.

## Beyond STEP

The value of $I$ is $\mathrm{e}\,E_1(1) \approx 0.5963$, where $E_1(t) = \int_t^\infty \mathrm{e}^{-u}u^{-1}\,\mathrm{d}u$ is the exponential integral. This number is known as the Gompertz constant. It cannot be written in terms of elementary functions, which is why the question only asks for bounds. Estimating integrals you cannot evaluate, by comparing them with ones you can, is a central skill in analysis (the "comparison test" for improper integrals). Abbott's *Understanding Analysis* develops it carefully.

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